Reader Report: Week 06

Probabilistic Invariance: VI-BIP and the Core Theorems

Daniel Hardesty Lewis (dl3645)

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Summary

Wu and Blei (2025) place a prior over binary selectors $z \in \{0,1\}^p$, where $z_j = 1$ indexes a feature in the invariant set S^* (Peters et al., 2016). The posterior $p(z \mid \mathcal{D}) \propto p(z) \prod_{e,i} \hat{g}(Y_i^e \mid X_{ei}^z)/\hat{p}_e(Y_i^e \mid X_{ei}^z)$ rewards selectors for which the pooled conditional $\hat{g}$ dominates each environment-specific conditional $\hat{p}_e$. Consistency (Thm 1) requires $\mu_{\min} > 0$, the minimum KL divergence between $\hat{g}$ and any $\hat{p}_e$ over non-invariant features (Wu and Blei, 2025, Def. 1). Contraction (Thm 2) gives $\mathrm{TV}(p(z \mid \mathcal{D}), \delta_{z^*}) = O(2^p\, e^{-\kappa n E \mu_{\min}})$. VI-BIP (Wu and Blei, 2025, Sec. 4) approximates the posterior via mean-field variational inference (Blei et al., 2017), optimizing the ELBO with the U2G gradient estimator.

Critique: What Does the Analyst Know About the Environments?

Peters et al. (2016), Fan et al. (2024), and Wu and Blei (2025) all enforce $P_e(Y \mid X_{S^*}) = \text{const}$ for all $e \in \mathcal{E}$, yet produce fundamentally different outputs: a confidence set, a point estimate, and a posterior distribution. The invariance condition alone does not determine the algorithm. What separates the three methods is what the analyst assumes about the environments before seeing data.

ICP assumes the least: environments are exchangeable, and adding more can only shrink the accepted set (Peters et al., 2016, Thm. 1). BIP inherits this agnosticism by weighting environments symmetrically in the product likelihood, but enumerates 2^p models to do so. Its contraction rate suggests an analogy with ICP's power: rearranging Thm 2 gives $nE \gtrsim O(p/\mu_{\min})$, where $\mu_{\min}$ plays a role similar to the signal strength governing ICP's rejection rate. The analogy is suggestive but informal, since $\mu_{\min}$ is a KL divergence while ICP's power depends on p-value thresholds under a different test statistic.

NegDRO breaks this symmetry. Its minimax problem $\min_b \max_{w \in U(\gamma)} \sum_e w_e \hat{\mathbb{E}}[(Y^e - b^\top X^e)^2]$ permits negative environment weights (Fan et al., 2024, Def. 1), penalizing the optimizer for performing well on a specific environment. This enforces risk parity across all pairs rather than robustness to the single worst one. NegDRO can therefore amplify a single harsh intervention when one environment is known *a priori* to be dominant. Neither paper acknowledges this as a structural incompatibility in their assumptions.

The benchmarking evidence inherits this bias. Wu and Blei (2025, Sec. 5) validate VI-BIP on 6,170 yeast genes across CRISPR knockouts where $\mu_{\min} > 0$ holds by construction, a setting of maximal practitioner knowledge. No evaluation is provided in observational or low-environment settings where $\mu_{\min}$ is unknown, which is where BIP's environment-agnostic design should matter most.

Questions

Both questions below concern whether the practitioner-knowledge gap between BIP and NegDRO can be bridged.

1. BIP's contraction guarantee requires $\mu_{\min} > 0$, but this quantity depends on the true invariant set z^*, which is what the method aims to discover. In the CRISPR benchmark, hard interventions guarantee $\mu_{\min} > 0$ by construction. In observational settings, the practitioner cannot verify this condition without already having solved the problem. Is there a testable proxy for $\mu_{\min} > 0$ that does not presuppose knowledge of z^*?
2. Wu et al. propose a size-restricted prior with parameter $p_{\max}$ to limit BIP's support to subsets of size at most $p_{\max}$. Could NegDRO's point estimate $\hat{\beta}^\gamma$ serve as the pre-screening filter, setting $z_j = 0$ for features with $|\hat{\beta}_j^\gamma| < \varepsilon$ to reduce the effective model count from 2^p to $2^{p'}$, and does this avoid the double-dipping problem that arises from using the same data to form both the prior and the likelihood?

References

D. M. Blei, A. Kucukelbir, and J. D. McAuliffe. Variational inference: A review for statisticians. *Journal of the American Statistical Association*, 112(518):859–877, 2017.

Y. Fan, S. Li, and Z. Yang. Causal invariance learning via efficient nonconvex optimization. *arXiv preprint arXiv:2412.11850*, 2024. URL https://arxiv.org/abs/2412.11850.

J. Peters, P. Bühlmann, and N. Meinshausen. Causal inference by using invariant prediction: identification and confidence intervals. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 78(5):947–1012, 2016.

Y. Wu and D. M. Blei. Bayesian invariance modeling of multi-environment data. *arXiv preprint arXiv:2506.22675*, 2025. URL https://arxiv.org/abs/2506.22675.